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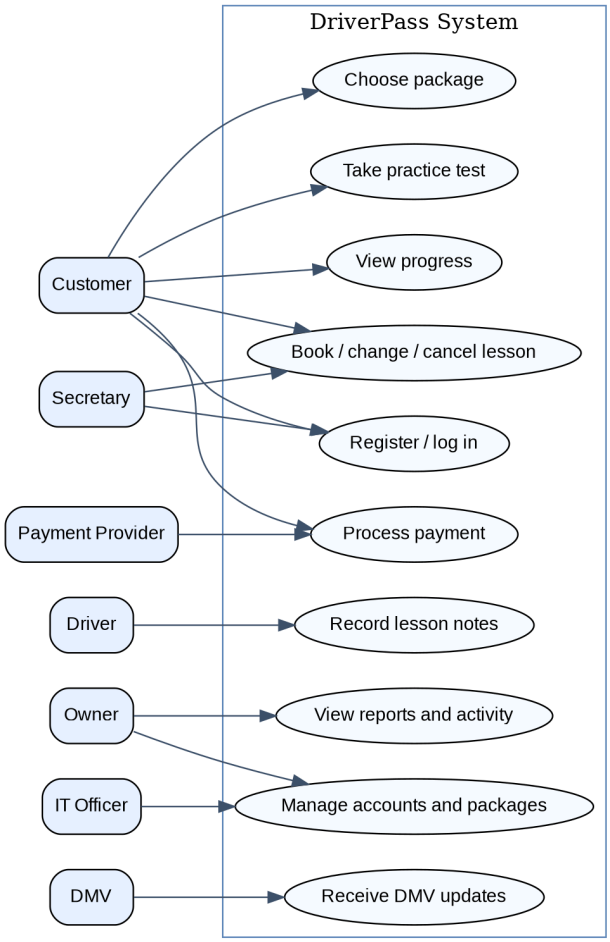
CS 255 System Design Document — DriverPass

Prepared by Deyanira Arroyo | CS 255 Project Two

UML Diagrams

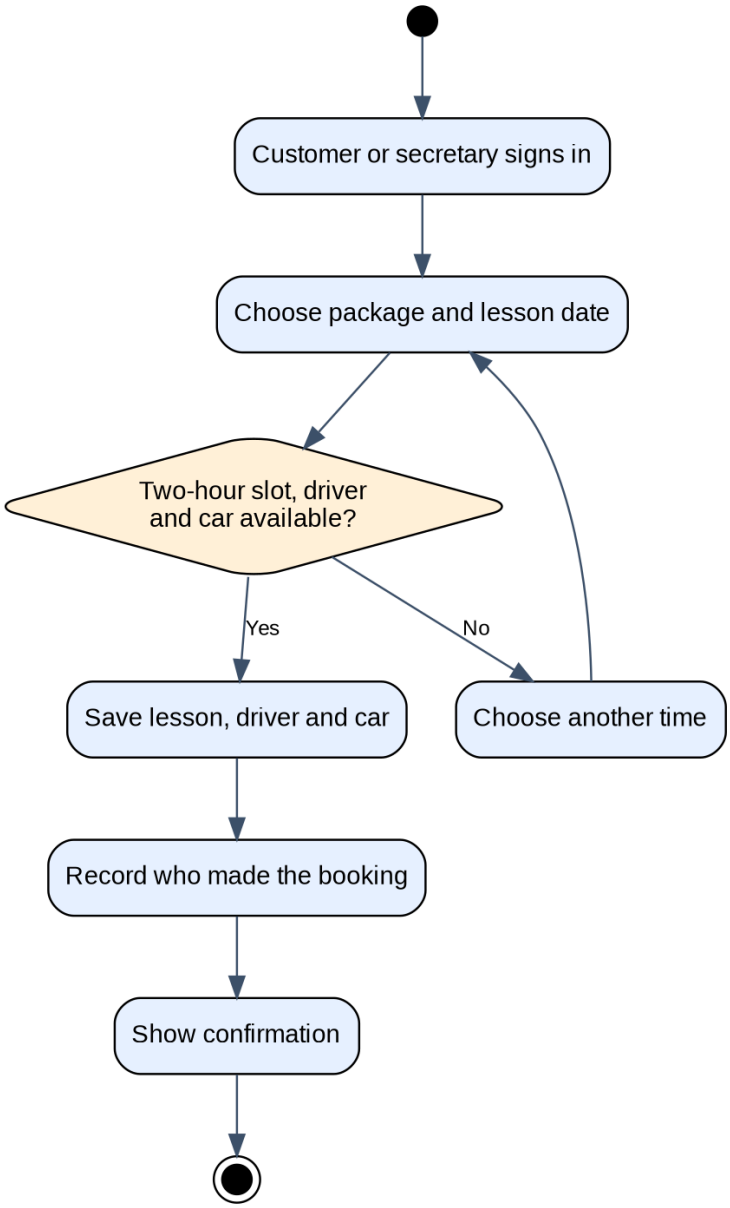
UML Use Case Diagram

The use case diagram shows how customers, the secretary, drivers, the owner, and the IT officer interact with DriverPass. The DMV and payment provider are outside services. It covers scheduling, training, reports, account management, and updates.

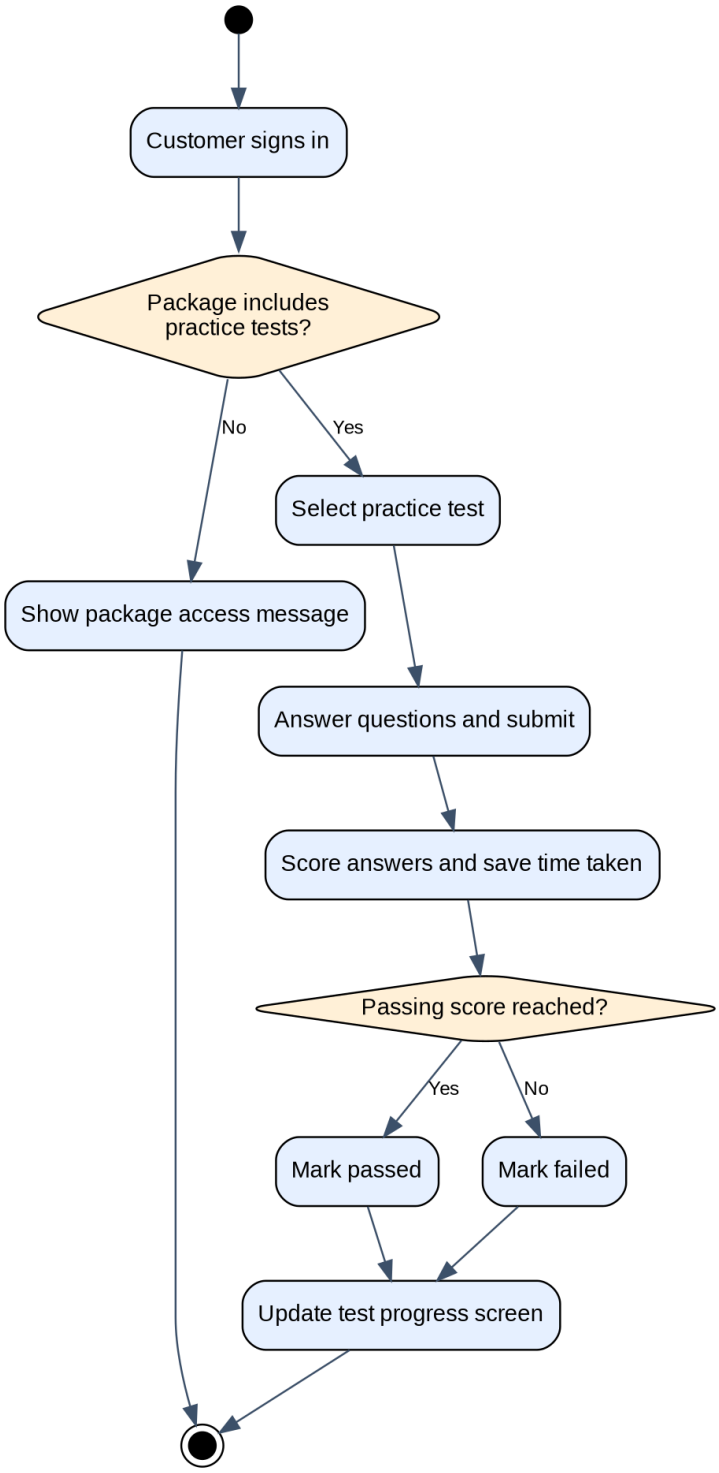


UML Activity Diagrams

Two use cases are shown below. The first follows a two-hour driving lesson booking and checks that a driver and car are free before saving it. The second follows an online practice test and checks whether the customer has access through their package.

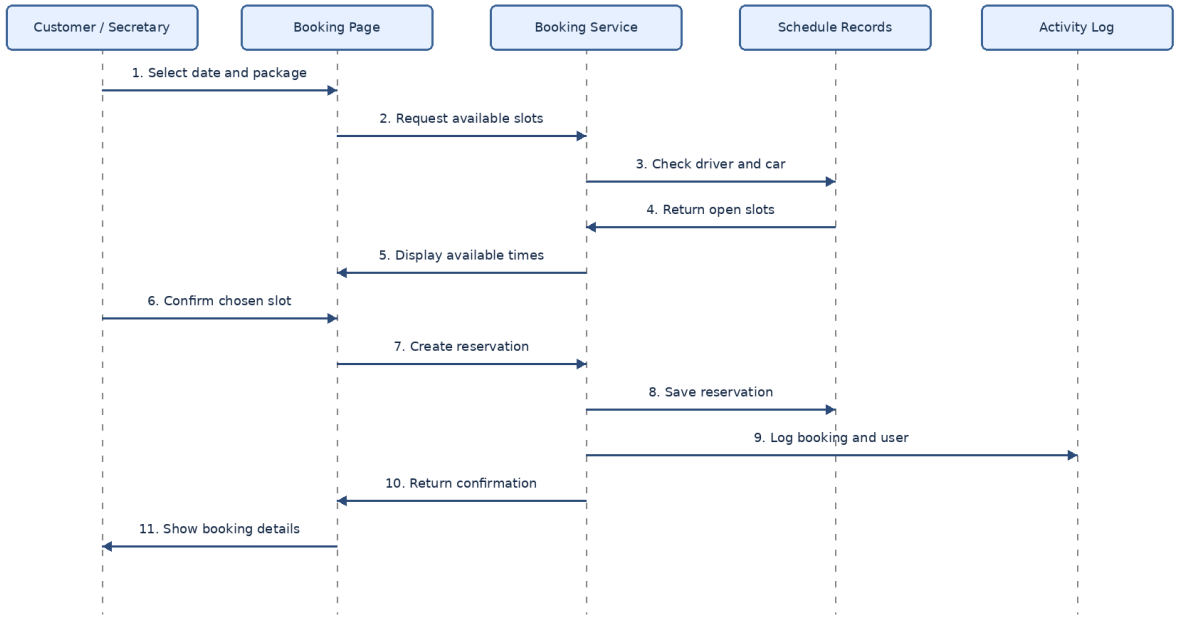


Activity Diagram 2 — Take a Practice Test



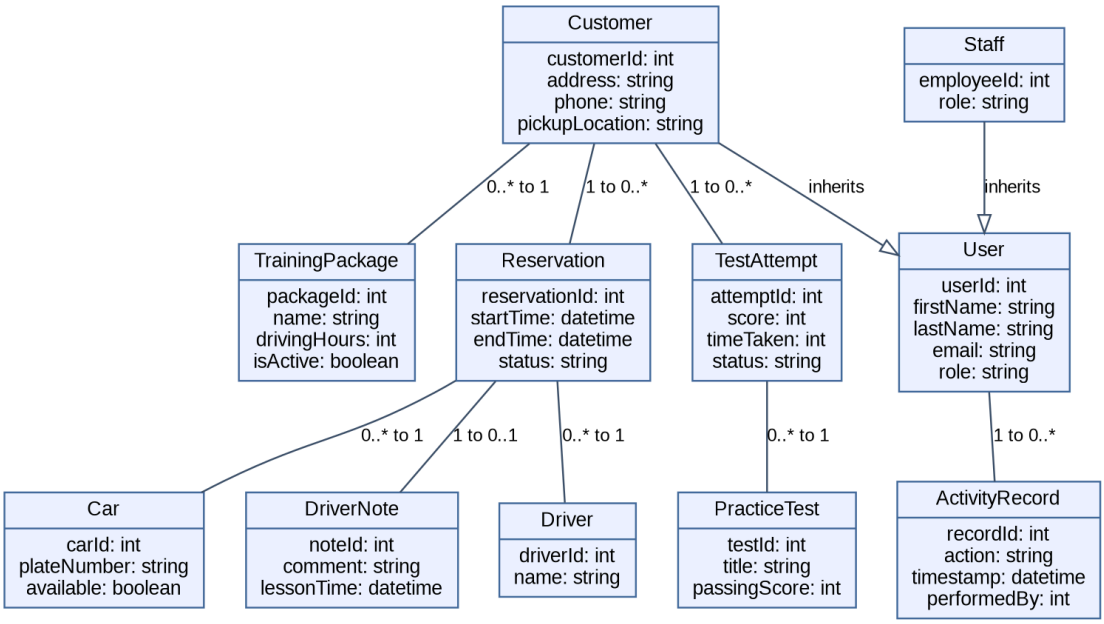
UML Sequence Diagram

This sequence follows the same booking use case as the first activity diagram. It shows the order of messages among the person booking, the booking page, the booking service, the schedule records, and the activity log.



UML Class Diagram

The class diagram identifies the main information DriverPass needs to store. A reservation connects a customer with a two-hour time slot, driver, and car. Test attempts keep progress, while activity records show who made changes. Attributes and multiplicities are included.



Technical Requirements

Hardware: Customers and employees need internet-capable computers, tablets, or smartphones with modern browsers. Office staff need a reliable internet connection and a printer for activity reports. No dedicated on-site server is required for the proposed cloud-based design.

Software: A responsive browser-based application will provide registration, package selection, two-hour lesson reservations, practice tests, driver notes, and reporting. The server will enforce permissions and scheduling rules. A relational database will store users, packages, drivers, cars, reservations, test results, and activity records. Reports should export to an Excel-compatible format.

Tools: Developers need a UML modeling tool such as Lucidchart, a code editor, version control, database tools, and automated testing. The team should test role permissions, booking conflicts, password resets, and mobile layouts before release.

Infrastructure: Cloud hosting should provide encrypted HTTPS connections, managed database storage, regular backups, monitoring, and recovery procedures. Passwords must be securely hashed. Access will be role-based: customers manage their own information, the secretary handles scheduling, drivers record lesson notes, Liam views reports, and Ian administers accounts. Reservation changes must be logged with the user and time. Payment information should be handled by a compliant payment service rather than storing card security codes.

Integration and constraints: DMV rule and question updates depend on a supported DMV data feed or agreed update process; the transcript does not confirm an available API. Changes to live records require an internet connection. The initial release can disable existing packages, but adding or removing training modules requires future development. Each driving session lasts two hours, and the booking service must prevent overlapping assignments of a driver or car.